Assignment – 1

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Part I: Short Answer Questions

Q1) Ans. A traditional file system stores data in separate files, which can lead to duplication and inconsistency. A DBMS stores data in a structured way and provides better security, consistency, and querying capabilities.

Example: A university managing student records is better off using a DBMS because multiple departments can access the same up-to-date data without maintaining separate files.

Q2) Ans. A strong entity has its own primary key and can be identified independently. A weak entity does not have a complete primary key of its own and depends on another entity for identification.

The primary key of a weak entity is formed by combining the primary key of the owner (strong) entity and the weak entity's partial key.

The relationship connecting them is called an identifying relationship.

Q3) Ans. Phone number should be modeled as a multivalued attribute because a student can have more than one phone number.

A simple attribute allows only one value, and a composite attribute is used when an attribute can be broken into smaller parts (e.g., address).

Q4) Ans. A super key is any set of attributes that uniquely identifies a tuple. A candidate key is a minimal super key (no attribute can be removed from it).

For Student(roll_no, email, name, branch):

Candidate Keys: {roll_no} , {email}

Super Keys that are not Candidate Keys: {roll_no, name} , {email, branch}

Q5) Ans.

- Violated Constraint**:** Foreign Key Constraint (Referential Integrity Constraint)
- It prevents references to records that do not exist in the parent table.
- The insert operation should be rejected by the DBMS.

Q6) Ans.

- Additional structure required**:** A separate relation (junction/bridge table) must be created.
- Primary key: Usually the combination of the primary keys of the participating entities.
- Why not as a column? Because one entity can be associated with multiple records of the other entity, which cannot be represented properly using a single column without causing redundancy and data anomalies.

Part II: Long Answer Question

Part A– Entity and Relationship Identification

Entities and Key Attributes

1. **Customer** : Customer(CustomerID, Name, Phone, Address)
2. **Restaurant** : Restaurant(RestaurantID, Name, Location)
3. **MenuItem** : MenuItem(ItemID, ItemName, Price)
4. **Order** : Order(OrderID, OrderDate, TotalAmount)

Relationships and Cardinalities :

1. Customer - Order (1:N) : A single customer can place many orders, but each order belongs to exactly one customer
2. Order - Restaurant (N:1) : Every order is tied to exactly one restaurant, but a restaurant can receive many orders over time.
3. Restaurant - MenuItem (1:N) : A restaurant offers many menu items, but each menu item belongs to one specific restaurant's menu.
4. Order - MenuItem (M:N) **:** One order can include multiple menu items, and the same menu item can appear in many different orders

Part B– Attribute Classification

Multivalued Attribute : Customer PhoneNumber

Reason: A customer may have more than one phone number, so multiple values can be stored for the same customer.

Composite Attribute : Customer Address

Reason: Address is a classic composite attribute because it's naturally made up of smaller meaningful parts - house number, street, city, state and pincode. Each sub-part has its own meaning and can be queried independently. Since the address consists of several meaningful subparts, it is a composite attribute.

Part C– Relational Schema Design

The M:N relationship between Order and MenuItem cannot be stored in either table directly, so a junction table is created.

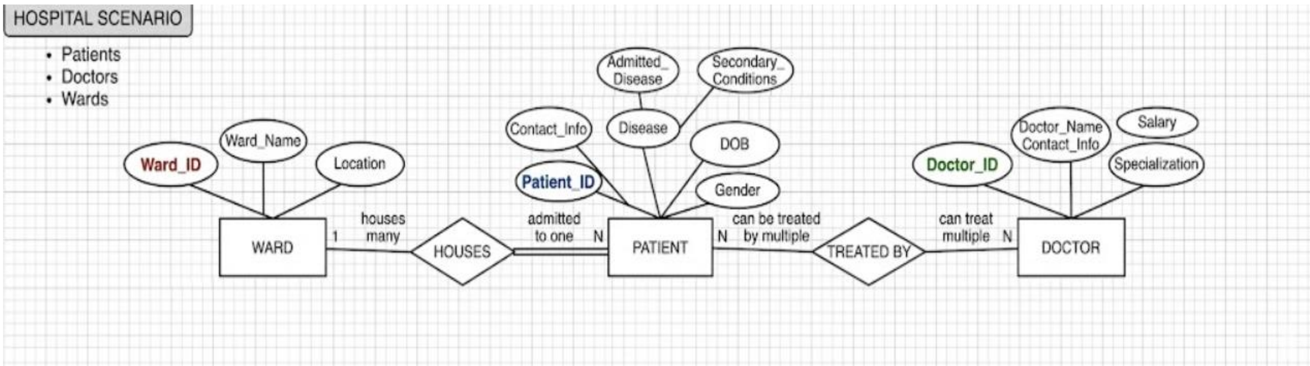
OrderItem(order_id, item_id, quantity)

Breakdown:

- **Primary Key:** (order_id, item_id) - composite, since the combination uniquely identifies each line item in an order
- **Foreign Key:** order_id - references Order(order_id)
- **Foreign Key:** item_id - references MenuItem(item_id)
- **Additional Attribute:** quantity - how many units of that item were ordered

Part III: ER Diagram Design

Question 1: ER diagram



Question 2 :

The Doctor-Patient relationship is many-to-many because a patient may receive treatment from several doctors during their stay, while a doctor can treat many different patients The composite primary key of this entity consists of DoctorID and PatientID, which uniquely identifies each doctor-patient treatment record.

For participation constraints, the Patient-Ward relationship has total participation on the Patient side because every admitted patient must be assigned to a ward. On the other hand, the Ward entity has partial participation since a ward can exist in the hospital even when no patients are currently admitted. In the Doctor-Patient relationship, both entities have partial participation because a doctor may be registered in the system before treating any patient, and a patient record may exist before treatment begins.

One design decision was to model Address as a composite attribute. An address can be divided into meaningful components such as street, city, state and PIN code. Storing these parts separately makes searching, filtering and updating address information much easier than storing it as a single text field.